

**17** Total number of moles of gas must be constant i.e.  $n = n_x + n_y$

$$\frac{pV}{RT} = \frac{p_x V_x}{RT} + \frac{p_y V_y}{RT}$$

$$pV = p_x V_x + p_y V_y$$

Answer: C

**18** The wave pulse undergoes a phase change of  $180^\circ$  or  $\pi$  radians on reflection when it